

# Curriculum Vitae

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## Jorge Alda Gallo

### Ph.D. in Theoretical Physics

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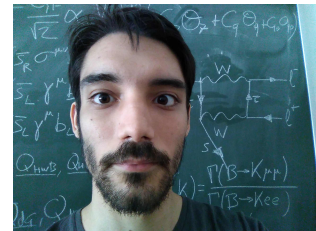
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🌐 <https://jorge-alda.github.io>

👤 Jorge-Alda

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## ☆ Research interests

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- New physics beyond the Standard Model.
- Flavour Physics.
- $B$ -meson anomalies.
- Effective Field Theories.
- Axions and Axion-like particles.
- Machine Learning applied to Physics.

## 🎓 Education

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B.Sc. in Physics, Universidad de Zaragoza

2011-2015

Average grade: 9.20/10. 13 Honours.

B.Ss. Project: “Cálculo numérico en teoría cuántica de campos de la materia condensada”.  
Under the supervision of David Zueco Láinez. Qualification: 9.5/10.

M.Sc. in Theoretical Physics, Universidad Complutense de Madrid  
2015-2016

Average grade: 9.34/10.

M.Sc. Project: *New Applications of the Coleman-Weinberg Model*. Under the supervision of J. A. Ruiz Cembranos. Qualification: 9.0/10.

Ph.D. in Physics, Universidad de Zaragoza 2016-2022

Under the supervision of Siannah Peñaranda Rivas.

Dissertation title: *A Glance into Flavour Physics with Effective Field Theories and Machine Learning*.

Defense date: 27th April 2022.

Qualification: *Cum Laude*.

Ph.D. School 2018

Taller de Altas Energías. Benasque (Huesca, Spain).

Ph.D. School 2022

WE-Heraeus Summer School SMEFT'22: Theory and Phenomenology of the Standard Model EFT. Siegen (Germany).



## Positions

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Universidad de Zaragoza April 2022 - February 2023

Università degli Studi di Padova October 2023 - September 2025

Università degli Studi di Padova December 2025 - December 2027

Contratto di Ricerca.



## Grants

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Grant JAE-Intro CSIC 2014

Project “*Caos semiclásico en sistemas de bosones con interacción*”, supervised by David Zueco Láinez.

CSIC-ICMA (Spanish National Research Council and Instituto de Ciencia de Materiales de Aragón).

PreDoc Grant, Diputación General de Aragón 2017-2022

Programa Ibercaja-CAI de Estancias de Investigación 2021

Grant No. CB 5/21.

Postdoctoral Scholarship for Extension of Studies Abroad for Life and Material Sciences, Ramón Areces Foundation 2023-2025

## Memberships

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CAPA 2019-present

Centro de Astropartículas y Física de Altas Energías. Zaragoza, Spain.  
capa.unizar.es

INFN 2023-2025

Associated to Istituto Nazionale di Fisica Nucleare, Sezione di Padova.

## Visiting Scholar Appointments

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Università degli Studi di Padova/INFN Summer 2021

Università degli Studi di Padova/INFN Summer 2023

University of California, San Diego September 2024

Laboratori Nazionali del Gran Sasso November 2024

Kavli IPMU, Tokyo May-June 2026

## Scientific Production

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**J. Alda, J. Guasch and S. Peñaranda: *Some results on Lepton Flavour Universality Violation***

Eur.Phys. J. C, 79 7 (2019) 588

doi:10.1140/epjc/s10052-019-7092-x

arXiv:1805.03636 [hep-ph]

**J. Alda, J. Guasch and S. Peñaranda: *Anomalies in B decays: A phenomenological approach***

Eur.Phys. J. Plus 137 (2022) 217

doi:10.1140/epjp/s13360-022-02405-3

arXiv:2012.14799 [hep-ph]

**J. Alda, J. Guasch and S. Peñaranda:** *Anomalies in B decays: Present status and future collider prospects*

arXiv:2105.05095 [hep-ph]

SLAC eConf C21-03-15.1

**J. Alda, J. Guasch and S. Peñaranda:** *Using Machine Learning techniques in phenomenological studies in flavour physics*

J. High Energ. Phys. 2022, 115 (2022)

doi:10.1007/JHEP07(2022)115

arXiv:2109.07405 [he-ph]

**J. Alda, J. Guasch and S. Peñaranda:** *Exploring B-physics anomalies at colliders*

arXiv:2110.12240 [hep-ph]

PoS(EPS-HEP2021)494

**J. Alda, A. W. M. Guerrero, S. Peñaranda and S. Rigolin:** *Leptonic Meson Decays into Invisible ALP*

Nucl.Phys.B 979 (2022) 115791

doi:10.1016/j.nuclphysb.2022.115791

arXiv:2111.02536 [hep-ph]

**J. Alda, G. Levati, P. Paradisi, S. Rigolin and N. Selimović:** *Collider and astrophysical signatures of light scalars with enhanced  $\tau$  couplings*

JHEP 06 (2025) 008

doi:10.1007/JHEP06(2025)008

arXiv:2407.18296 [hep-ph]

**J. Alda, A. Mir and S. Peñaranda:** *Flavour anomalies and Machine Learning: an improved analysis*

arXiv:2412.15830 [hep-ph]

Submitted to EJPC

**J. Alda, C. Brogini, G. Di Carlo, L. Di Luzio, D. Piatti, S. Rigolin and C. Toni:** *Time modulation of weak nuclear decays as a probe of axion dark matter*

Phys.Rev.D 111 (2025) 3, 035022

doi:10.1103/PhysRevD.111.035022

arXiv:2412.20932 [hep-ph]

**J. Alda, M. Fuentes Zamoro, L. Merlo, X. Ponce Díaz and S. Rigolin:** *Comprehensive ALP Searches in Meson Decays*

arXiv:2507.19578 [hep-ph]

Submitted to JHEP

**J. Alda, M. Fuentes Zamoro, L. Merlo, X. Ponce Díaz and S. Rigolin: *Comprehensive ALP Searches in Meson Decays***  
 arXiv:2508.08354 [hep-ph]  
 Submitted to Computer Physics Communications

**A. Mir, J. Alda and S. Peñaranda: *B-Meson Anomalies: Effective Field Theory Meets Machine Learning***  
 arXiv:2510.17742 [hep-ph]  
 Contribution to EPS-HEP2025

**M. Fuentes Zamoro, J. Alda, L. Merlo, X. Ponce Díaz and S. Rigolin: *Tackling ALP searches in meson decays with ALPaca: a phenomenological approach***  
 Proceedings of 3rd General Meeting of the COST Action COSMIC WISPer (CA21106)  
<https://pos.sissa.it/507/057>

**J. Alda, L. C. Bresciani, G. Levati, P. Paradisi, S. Rigolin and N. Selimović: *Leptonic Couplings of Axion-Like Particles: Constraints from Present and Next-Generation Colliders***  
 In preparation

**J. Alda, S. Rigolin: *ALPs in Chiral Perturbation Theory: general framework and baryon interactions***  
 In preparation

**J. Alda: *Lecture notes on Machine Learning applications for global fits***  
 In preparation

**J. Alda, J. Asorey, A. Mir, S. Peñaranda: *Efficient Likelihood Emulation for High-Energy Physics and Cosmology Global Fits***  
 In preparation



## Talks and conferences

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**2nd Red LHC Workshop. Madrid, Spain. 9-11 May 2018**  
 Talk “Some Results on Lepton Flavour Violation”.

**Taller de Altas Energías. Benasque (Huesca, Spain) 2-15 September 2018**  
 Talk “Some Results on Lepton Flavour Violation”.

**X CPAN Days. Salamanca, Spain. 29-31 October 2018**  
 Talk “Complex Wilson coefficients in the analysis of  $B$ -anomalies”.

**I Jornadas de Jóvenes Investigadores CAPA. Zaragoza, Spain. 7 May 2019**  
 Invited Talk “Effective Theories for  $B$ -meson anomalies”.

**I Jornadas del Programa de Doctorado de Física. Zaragoza, Spain. 20 June 2019**  
 Invited Talk “Effective Theories for  $B$ -meson anomalies”.

**XXXVII Bienal de Física de la Real Sociedad Española de Física. Zaragoza, Spain. 15-19 de July 2019**

Talk “Some Results on Lepton Flavour Universality Violation”.

**International Workshop on Future Linear Colliders - LCWS2021. Online. 15-18 March 2021**

Talk “Anomalies in  $B$  mesons decays: Present status and future collider prospects”.

**European Physical Society Conference on High Energy Physics 2021 (EPS-HEP2021). Online. 26-30 July 2021**

Poster “Exploring B-physics anomalies at colliders”.

**Seminar, Department of Theoretical Physics. University of Zaragoza, Spain. 18 November 2021**

Invited Talk “Leptonic Mesons Decays into invisible ALP”.

**II Jornadas del Programa de Doctorado de Física. University of Zaragoza, Spain. 3 December 2021**

Invited Talk “Leptonic Mesons Decays into invisible ALP”.

**Seminar, Department of Theoretical Physics. University of Zaragoza, Spain. 20 January 2022**

Invited Talk “Using Machine Learning techniques in phenomenological studies in flavour physics”.

**Seminar, Institute of Theoretical Physics (IFT). Autonomous University of Madrid, Spain. 27 January 2022**

Invited Talk “Using Machine Learning techniques in phenomenological studies in flavour physics”.

**XIII CPAN Days. Huelva, Spain. 21-23 March 2022**

Talk “Using Machine Learning techniques in phenomenological studies in flavour physics”.

**5th Inter-experiment Machine Learning Workshop. CERN, Switzerland. 8-13 May 2022**

Talk “Using Machine Learning techniques in phenomenological studies in flavour physics”.

**Saturnalia Workshop. University de Zaragoza/CAPA, Spain. 21 December 2022**

Invited Talk “Using Machine Learning techniques in phenomenological studies in flavour physics”.

**Saturnalia Workshop. University de Zaragoza/CAPA, Spain. 20 December 2023**

Invited Talk “Light New Physics coupling to  $\tau$ ”.

**Seminar, Institute of Theoretical Physics (IFT). Autonomous University of Madrid, Spain. 06 June 2024**

Invited Talk “Leptophilic Axion-Like Particles”.

**Seminar, University of California, San Diego. 17 September 2024**

Invited Talk “Signatures of light scalars with enhanced  $\tau$  couplings”.

**Flavour for New Physics at Present and Future Colliders Workshop. Mainz Institute for Theoretical Physics, Johannes Gutenberg University, Germany. 19 June 2025**

Invited Talk “Present and future of ALP searches in colliders”.

**Invisibles25 Workshop. CERN, Switzerland. 4 September 2025**

Talk “ALP searches in meson decays with ALPaca: a phenomenological approach”.

**Implications of LHCb measurements and future prospects. CERN, Switzerland. 5 November 2025**

Invited Talk “ALP searches in meson decays with ALPaca”.

**Saturnalia Workshop. University de Zaragoza/CAPA, Spain. 18 December 2025**

Invited Talk “GeV-ALP searches with ALP-aca”.

**ALPs across scales. Mainz Institute for Theoretical Physics, Johannes Gutenberg University, Germany. 23 June 2026**

Talk “New insights in ALP interactions with hadrons”.



## Contributions to public repositories

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flavio

1 pull request merged: <https://github.com/flav-io/flavio/pull/160>

smelli

1 pull request: <https://github.com/smelli/smelli/pull/45>

ALP Automatic Computing Algorithm (ALP-aca)

Main contributor. <https://github.com/alp-aca>



## Teaching

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September 2019

**Ph.D. School “Taller de Altas Energías de Benasque” (Huesca, Spain).**

Associate teacher.

2019-2020

**Differential Equations**

Problem-solving sessions, 38 teaching hours.

Second year course, Bachelor Degree in Physics, Universidad de Zaragoza.

**General Physics**

Laboratory sessions, 10 teaching hours.

First year course, Bachelor Degree in Mathematics, Universidad de Zaragoza.

2020-2021

**Differential Equations**

Problem-solving sessions, 38 teaching hours.

Second year course, Bachelor Degree in Physics, Universidad de Zaragoza.

**General Physics**

Laboratory sessions, 10 teaching hours.

First year course, Bachelor Degree in Mathematics, Universidad de Zaragoza.

2021-2022

**Differential Equations**

Problem-solving sessions, 38 teaching hours.

Second year course, Bachelor Degree in Physics, Universidad de Zaragoza.

**Co-direction of B.Sc. Project: Alejandro Mir**

10 teaching hours. Fourth year course, Bachelor Degree in Physics, Universidad de Zaragoza.

**General Physics**

Laboratory sessions, 10 teaching hours.

First year course, Bachelor Degree in Mathematics, Universidad de Zaragoza.

2022-23

**Co-direction of M.Sc. Project: Alejandro Mir**

10 teaching hours. Master Degree in Physics, Universidad de Zaragoza.

**Co-direction of B.Sc. Project: Francisco J. Gómez-Fauro**

10 teaching hours. Fourth year course, Bachelor Degree in Physics, Universidad de Zaragoza.

2024-present

**Co-direction of Ph.D. Thesis: Alejandro Mir**

Co-direction of Ph.D. Thesis in Physics, Universidad de Zaragoza.





## Scientific Dissemination

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**Dark Matter Day. University of Zaragoza, Spain. 31 October 2022**

**Organizing Committee of the 2022 Saturnalia Workshop. University of Zaragoza/CAPA, Spain. 16-22 December 2022**

<https://indico.capa.unizar.es/event/32/>

**Local Organizing Committee of PLANCK2025 - The 27th International Conference From the Planck Scale to the Electroweak Scale. Padova, Italy. 26-30 May 2025**

<https://indico.dfa.unipd.it/event/1200/>

**4th Training School on Computing Challenges, COMCHA network. University of Zaragoza/CAPA, Spain. 8-15 April 2026**

<https://indico.capa.unizar.es/event/42/>

Topical lecture and hands-on session “ML for global fits”.



## Languages

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|         |                          |
|---------|--------------------------|
| Spanish | ■■■■■                    |
| English | ■■■■■                    |
| Italian | ■■■■■ [C1 certification] |
| German  | ■■■■■                    |
| French  | ■■■■■                    |



## Programming languages

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|                                                                              |       |
|------------------------------------------------------------------------------|-------|
| $\text{T}_{\text{E}}\text{X}/\text{L}_{\text{A}}\text{T}_{\text{E}}\text{X}$ | ■■■■■ |
| Python                                                                       | ■■■■■ |
| C/C++                                                                        | ■■■■■ |
| Mathematica                                                                  | ■■■■■ |
| Docker                                                                       | ■■■■■ |
| Rust                                                                         | ■■■■■ |



## Awards

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XXII Spanish Physics Olympiad

2011

Silver Medal (Rank 15).

Second position in the regional Aragonese phase.

20th International Mathematics Competition

2013

Bronze Medal (rank 177).

## About this CV

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This CV was last updated in March 12, 2026.  
You can find the latest version at  
[https://github.com/Jorge-Alda/Jorge-Alda/  
releases/latest/download/CV.pdf](https://github.com/Jorge-Alda/Jorge-Alda/releases/latest/download/CV.pdf)

